

3rd class

N.B pin mode set always Hexadecimal use always!

Ex:

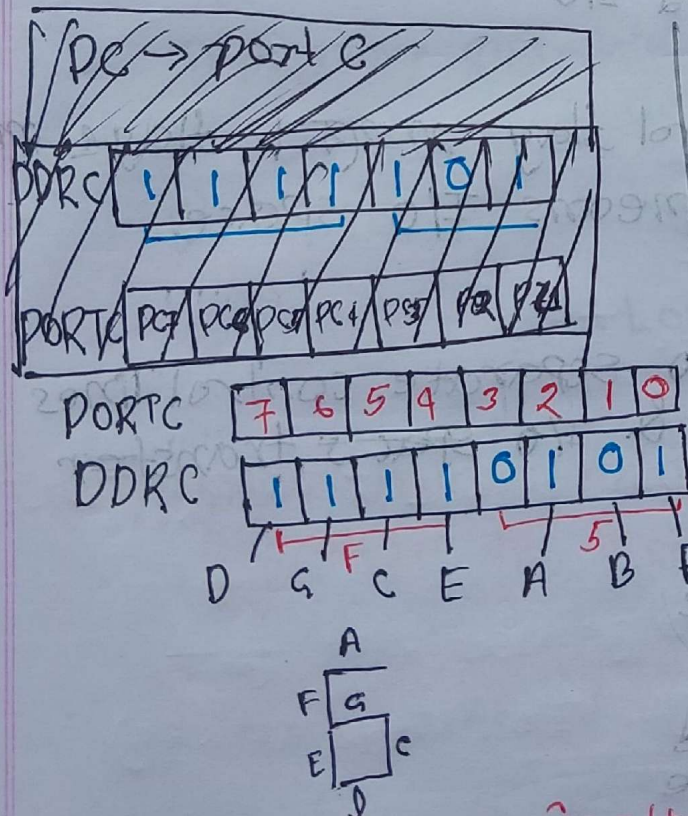
If an 8-bit data-direction Register contains 39 H (39 Hexa), then which pin's of the port is set as input and output.

$$\begin{aligned} \text{DDR}_x &: 39 \text{ H} \\ &= 00110100 \end{aligned}$$

Output pin's: 2, 4, 5

Input pin's: 0, 1, 3, 6, 7

Summer 25 81 (b)

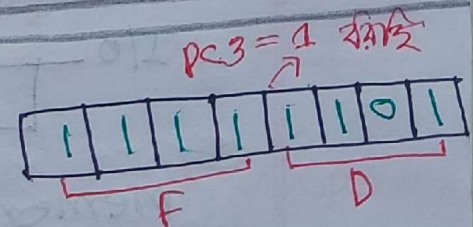


PC means PORTC

B → input PC1

Others → output

PC3 → not connected to anything (so 0/1 means or किसी भी)



Hexa: FD H

means Hexa

6 pin's (D, C, B, A, E, F) B pin is off means
Hexa: F5 H

Why? : 1 means output
0 " input

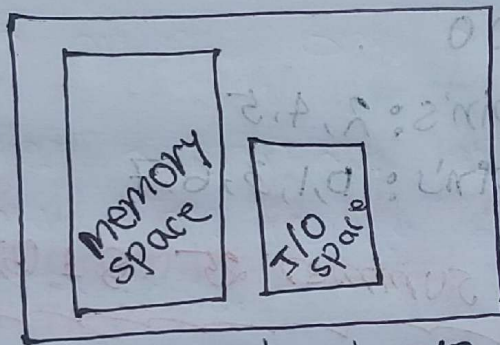
programmed i/o

I/O ports are two types based on the address:

1) Standard i/o

Two separate address spaces are used one for memory location (RAM) and other for I/O device.

Ex:



standard I/O

Flag/control flag means RAM, Flag=0 means I/O space.

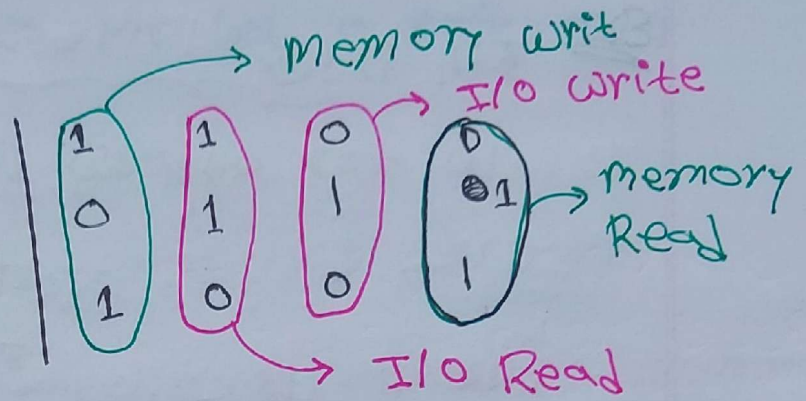
There are two separate control lines for memory space & I/O space transfer

I/O → read
→ write

memory → read
→ write

Control Bus:

- i) I/O $R/\bar{W}$
- ii) memory $R/\bar{W}$
- iii) $M/\bar{I/O}$ pin



N.B mainly 3 or Bus control ব্যবস্থা যা করে কি operation করতে।

~~i) $I/O R/\bar{W} = 1$ means $\Rightarrow I/O I/O$~~

i) $I/O R/\bar{W} = \text{high} \Rightarrow I/O 1/0 \Rightarrow I/O \text{ Read}$

~~ii) $\text{memory } R/\bar{W} = \text{high}$~~
 $= \text{Low} \Rightarrow I/O 0/1 \Rightarrow I/O \text{ write}$

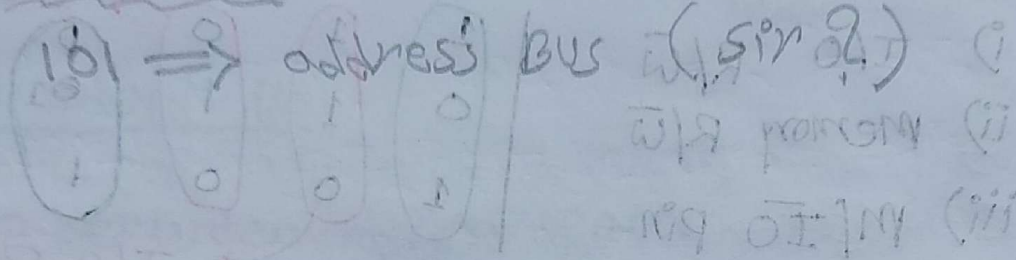
ii) $\text{memory } R/\bar{W} = \text{high} \Rightarrow \text{memory } 1/0 \Rightarrow \text{memory Read}$
 $= \text{Low} \Rightarrow \text{memory } 0/1 \Rightarrow \text{memory write}$

iii) $M/\bar{I/O}$ pin = high $\Rightarrow 1/0 \Rightarrow \text{memory operation on going}$
 $= \text{Low} \Rightarrow 0/1 \Rightarrow I/O \text{ operation is on going}$

N.B i) প্রথম দৈর্ঘ্য $M/\bar{I/O}$ pin কি, যদি memory operation বা then দৈর্ঘ্য memory $R/\bar{W}$ কি হবে।

ii) প্রথম দৈর্ঘ্য $M/\bar{I/O}$ pin কি, যদি I/O operation বা then দৈর্ঘ্য I/O $R/\bar{W}$ কি হবে।

Ex: 5th Index



Disadvantage:

- i) need one extra control bit to define I/O or memory operation.
- ii) After operation ~~the~~ programmed don't show ~~the~~ on display and store the result at a time.

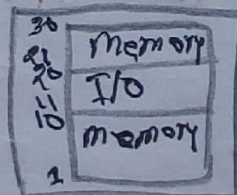
Advantage:

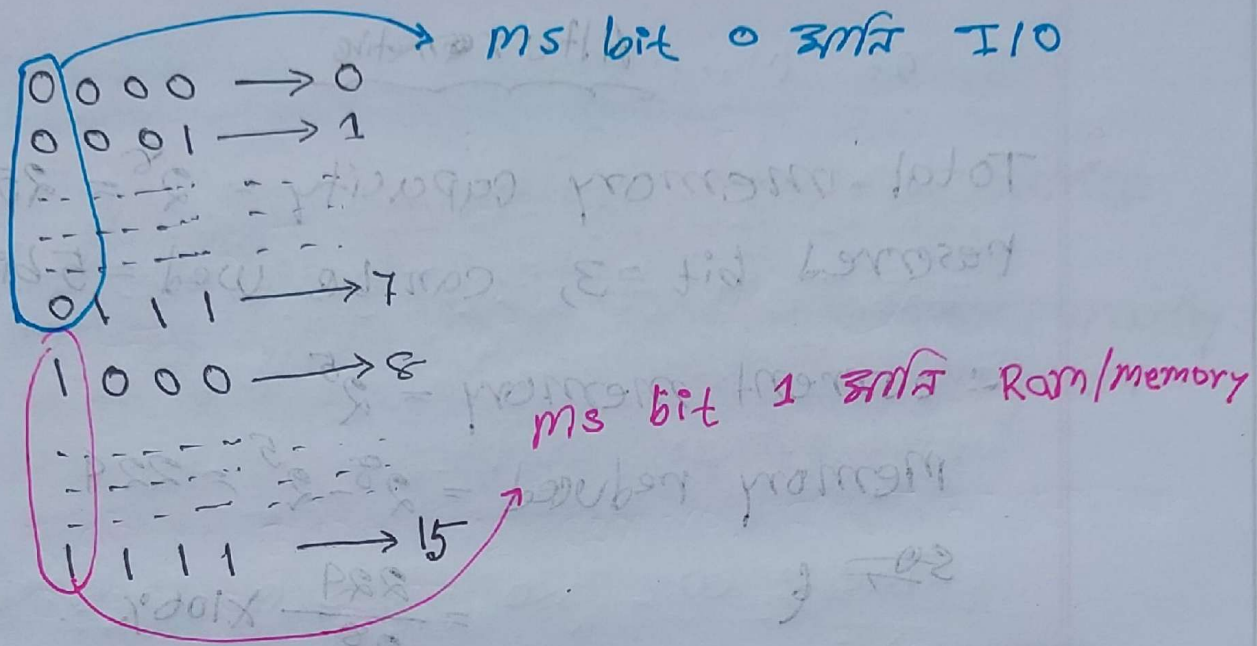
If there is no restriction for memory then it's better.

ii) Memory mapped I/O:

only ~~single~~ address space ~~is~~ ~~shared~~ ~~for~~ ~~memory~~ ~~and~~ ~~I/O~~.

Space Range ~~আকার~~ use ~~করে~~, ~~যেমন~~
 Rom (index 1...10), I/O (index 11...20)





Advantage:

মডি বাল দি faster আশে দি Better.

Disadvantage:

বিদ্রো অনেক Waste হয় কারণ Range আশে define আশে।

Full 25 (9 16)

$\underbrace{11111111}_3 \underbrace{11111111}_5] 8 \text{ bit address bus}$

Total indices: 2^8

Available indices for storage: 2^5

Reduction: $2^8 - 2^5$

$$= \frac{2^8 - 2^5}{2^8} \times 100\%$$

$$= 87.5\%$$

Alternative

Total memory capacity = $2^8 = 256$

reserved bit = 3, can be used = 5 bit

$\therefore$ current memory = 2^5

Memory reduced = $2^8 - 2^5 = 224$

~~50~~ 6

$$= \frac{224}{2^8} \times 100\%$$

= 87.5% Waste.

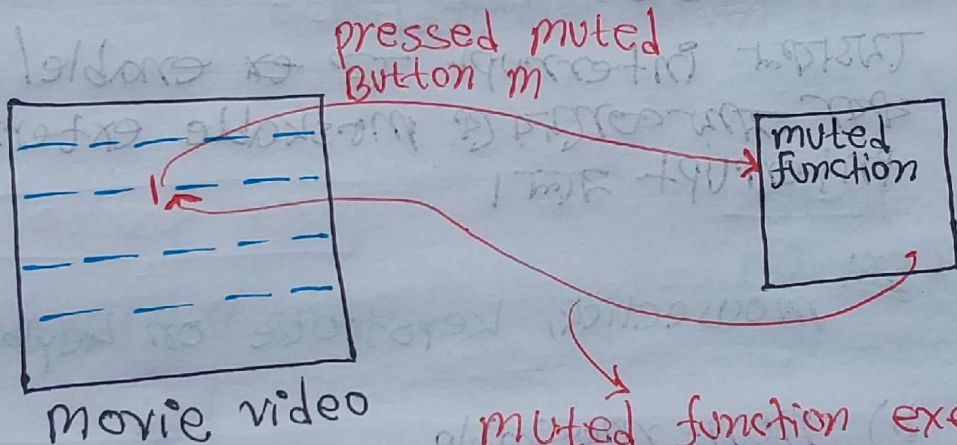
જિદ્વાસ ત્રાદિ સત્તજ ડાઉ % used રહેલે તારલ

$$\begin{aligned} \text{રૂપી} &= \frac{2^5}{2^8} \times 100\% \\ &= 12.5\% \end{aligned}$$

ii) Interrupt driven I/O:

Interrupt I/O, an ~~example~~ external device can force the microprocessor to stop executing the current program temporarily so that it can execute another program known as an interrupt service.

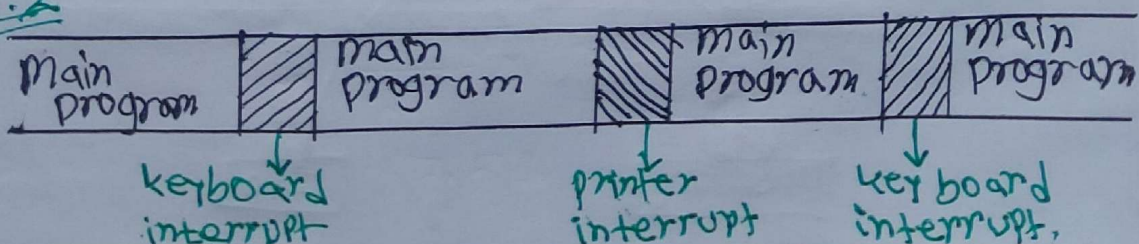
Ex: 1



muted function execute করার
পর movie টি interrupt point
থেকে play হবে।

Mainly ~~বিশেষ~~ ~~বর্ত~~ movie চলছিলো, ~~সহ~~ ~~সহ~~
Use m Button ~~চালছে~~ movie ~~pause~~ বা Pause করে
muted function টি execute করতে CPU, then movie
টি ~~মুদ্রা~~ ~~থেকে~~ pause হইছে (স্থান থেকে চলছে)

Ex: 2



Types of interrupt driven i/o

i). External interrupt:

যদি CPU টি বাহ্যিক generate করে তবে
External interrupt

Ex: mouse, keyboard

There are two types of External interrupt

a) Maskable

যদি interrupt টি enable/disable
করা যায় তবে maskable external
interrupt বলে।

Ex: mouseclick, keystroke on keyboard

b) Non-maskable

যদি interrupt টি enable করা
করা যায় না তবে non-maskable
external interrupt বলে।

Ex: power failure-sensing circuit

→ current interruption CPU টি interrupt
থাকবে backup করা হবে

ii) internal interrupt:

Internal interrupt CPU চি- ডিভাইস generate
এই আদেই internal interrupt বলে।

Ex:

$$a = 0$$

$$b = 5$$

$$c = b/a$$

} ~~error~~ CPU code execute
for error মানে!

interrupt driven I/O

